Statistical Learning

Framework, Prediction, and Inference

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Roadmap

1. What Is Statistical Learning?

1.1 The Statistical Learning Problem

Statistical learning (also called machine learning) is a set of tools for understanding data and making predictions from data.

The starting point is always the same:

- We have some observed data — pairs of inputs X and outputs Y.
- There is some unknown relationship between X and Y that we want to learn.
- Once we have learned that relationship, we can either predict new outcomes or understand how X is associated with Y.

The rest of this lecture builds the framework that makes these vague ideas precise. The rest of the course uses that framework to build specific methods.

2. Three Motivating Examples

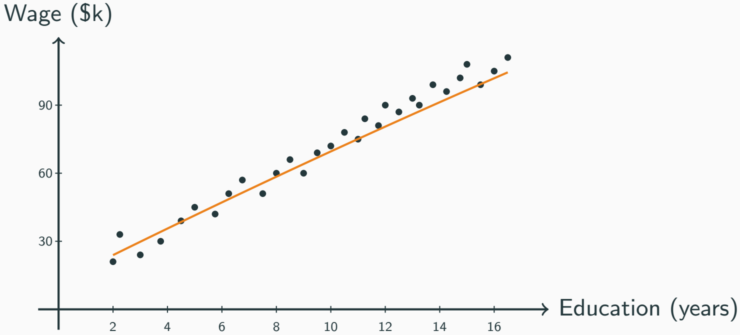
2.1 Example: Wages

Setup. We have data on $n = 3{,}000$ U.S. workers — for each, we know their years of education, age, and annual wage.

Questions.

- Predict. For a 27-year-old with a bachelor's degree, what wage should we expect?
- Understand. How do expected wages vary with education, holding age fixed?

2.1 Example: Wages 2



Y is the wage (a continuous number). X is education and age (a vector). The orange curve — the relationship we want to learn — is what we'll call f.

2.2 Example: Stock Market Direction

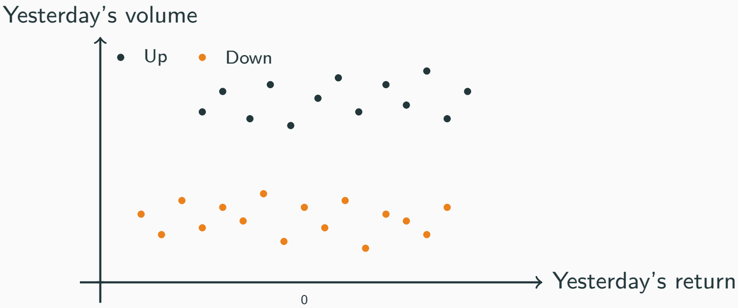
Setup. We have daily data on the S&P 500 index over several years — for each day, we know yesterday's return, yesterday's trading volume, and whether the index went up or down today.

Questions.

- Predict. Given yesterday's return and volume, will the index go up tomorrow?
- Understand. Does high trading volume signal future direction?

Here Y is *not* a continuous number — it takes only two values, "Up" or "Down." This is a different kind of problem (we'll formalize it in §5). X is yesterday's return and volume.

2.2 Example: Stock Market Direction 2

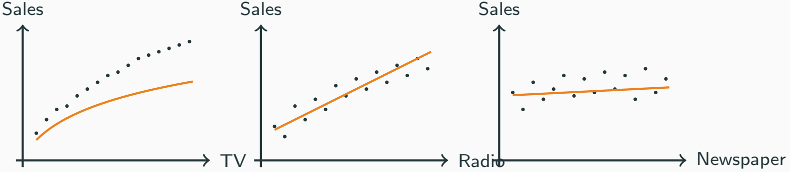


2.3 Example: Advertising and Sales

Setup. A company sells a product in 200 cities. For each city, we have TV ad spending, radio ad spending, newspaper ad spending, and sales.

Questions.

- Predict. If we spend \$50K on TV and \$20K on radio in a new city, what sales should we expect?
- Understand. Which channels are most strongly associated with sales? Is newspaper spending predictively useful?



2.3 Example: Advertising and Sales 2

Y is sales (continuous). X has *three* components (X_1, X_2, X_3 for the three channels). Notice TV and radio show a strong relationship with sales; newspaper looks almost flat.

3. Inputs and Outputs

3.1 The Two Roles

In every example above, we have two roles for variables:

Definition

- Input variables (also called predictors, features, or independent variables): the information we observe and want to use. Denoted X (or $X_1, X_2, \ldots, X_p$ when there are multiple).
- Output variable (also called response or dependent variable): the quantity we want to predict or understand. Denoted Y.

Example	Inputs X	Output Y
Wages	education, age	annual wage
Stocks	yesterday's return, volume	up or down
Ads	TV, radio, newspaper spend	sales

Notation reminder. We use p for the number of predictors. So the wage example has $p = 2$; the advertising example has $p = 3$.

4. The General Framework

4.1 The Statistical Learning Model

We assume the outputs and inputs are related by:

Key Result — The model

$$\boxed{Y \;=\; f(X) \;+\; \epsilon}$$

where:

- f is a fixed but unknown function of X that captures the systematic relationship.
- Within one observation, ϵ is a random error term with $\mathbb{E}[\epsilon \mid X] = 0$: the predictor does not systematically predict whether the error is positive or negative.

Everything in this course can be read as "different ways of approximating f from data, given that the data is noisy."

Notation reminder. $f : \mathbb{R}^p \to \mathbb{R}$ is a function — given a value of X, it returns a single number $f(X)$ (recall Lecture 1 §6.1). This f is what we estimate.

4.2 What Is f?

f is the systematic part of the relationship between X and Y — the piece that's the same for every observation with the same X.

We don't know what f looks like. It might be:

- A straight line: $f(X) = \beta_0 + \beta_1 X$.
- A polynomial: $f(X) = \beta_0 + \beta_1 X + \beta_2 X^2$.
- Something more flexible: $f(X) = \beta_0 \log(X + 1)$, or a step function, or a neural-net composition...

Our job is to learn f from the data. Different methods make different assumptions about its shape:

- Linear methods (Lectures 6–9) assume f is a linear function of X.
- Non-parametric methods (Lecture 11) impose much weaker assumptions.
- Tree-based methods (Lecture 14) build f as a piecewise constant function on rectangular regions.

4.3 What Is ϵ?

ϵ is the noise — whatever piece of Y is *not* captured by X through f. The main ideas are:

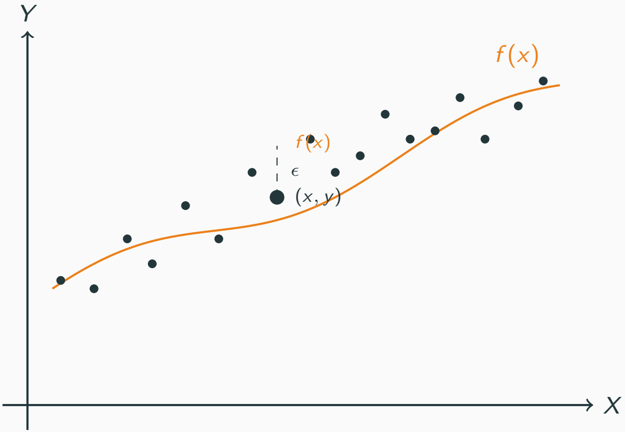
- Within observation i, the random error satisfies $\mathbb{E}[\epsilon_i \mid X_i] = 0$: positive and negative errors balance out at each predictor value.
- Across observations, the pairs (X_i, Y_i) are independent.
- The overall variance of ϵ is denoted σ^2.

Why is ϵ there?

- Variables we didn't measure but that affect Y.
- Inherent randomness in the system.
- Measurement error.

In short: ϵ is the part of Y that X does not explain.

4.4 Picture: $Y = f(X) + \epsilon$



4.4 Picture: $Y = f(X) + \epsilon$ 2

The orange curve is f — the systematic relationship. The dark dots are the observed pairs (x, y). For each observation, the vertical gap between the dot and the curve is one realization of ϵ.

Our job: use the dots (the data) to learn the curve (f).

4.5 The Probabilistic Perspective

Why does this course start with probability and statistics? Because the model $Y = f(X) + \epsilon$ is fundamentally a random object. Once you accept that:

- ϵ is a random variable (Lectures 1–2).
- Therefore Y is a random variable for any fixed X.
- Our estimate $\hat{f}$ is itself *random*, because it's built from a noisy training sample.
- Therefore predictions $\hat{Y} = \hat{f}(X)$ are random too.

To make any quantitative statement — "how accurate is this prediction?", "how strongly is X_1 associated with Y?" — we need the language of probability:

4.5 The Probabilistic Perspective 2

- Expectation, variance, independence (Lectures 1–2).
- Sample mean and the CLT (Lecture 3) for the behavior of estimators built from data.
- Confidence intervals and hypothesis tests (Lecture 4) for inference about f.

Without this machinery, "my model predicts \$52K" is meaningless. With it, we can say "\$52K $\pm$ \$3K, with 95% confidence."

5. Regression vs Classification

5.1 Two Kinds of Output

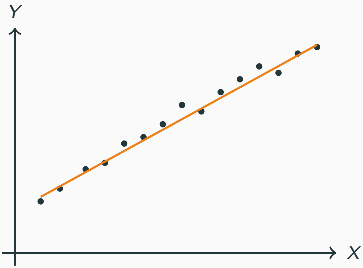
The framework $Y = f(X) + \epsilon$ covers two genuinely different problem types — distinguished by what kind of variable Y is.

Definitions

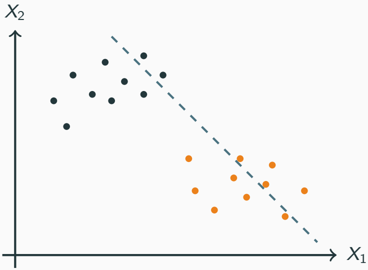
- Regression: Y is quantitative (a continuous number on $\mathbb{R}$). Examples: wage, sales, blood pressure.
- Classification: Y is qualitative (a category from a finite set). Examples: up/down, spam/not spam, A/B/C/D.

Most of this course is built around this distinction. The math differs depending on which kind of Y you have.

5.2 Side-by-Side Picture



Regression
Y is quantitative



Classification
Y is qualitative

Regression (left): $\hat{f}$ is a curve that maps X to a number $\hat{Y}$.

Classification (right): $\hat{f}$ is a *boundary* that splits the X-space into regions, one per class. Predicting means picking which side of the boundary you're on.

5.3 Where We'll Cover Each

Regression methods in this course

- Linear regression (Lectures 6–9)
- Shrinkage / regularization — ridge, lasso (Lecture 10)
- Non-parametric regression (Lecture 11)

Classification methods

- Logistic regression (Lecture 12)
- Random forests (Lecture 14)

Model assessment for *both* problem types uses Lecture 13 resampling methods (cross-validation and bootstrap).

6. Supervised vs Unsupervised

6.1 Two Settings for Learning

Whether we can fit the model $Y = f(X) + \epsilon$ depends on what data we have.

Definitions

- Supervised learning: we observe both inputs X *and* outputs Y in the training data. We use this labeled data to learn f, then predict Y for new X.
- Unsupervised learning: we observe only inputs X (no Y). Goal: discover *structure* in X — groups of similar observations, low-dimensional summaries, anomalies.

Examples:

- Supervised: wages (we have $Y =$ wage); stocks (we have $Y =$ up/down); ads (we have $Y =$ sales). All of our examples so far.

6.1 Two Settings for Learning 2

- Unsupervised: customer segmentation from purchase histories (group similar shoppers); topic discovery from documents; anomaly detection.

This course covers supervised learning only. The rest of the lecture assumes we have (X, Y) pairs.

7. Why Estimate f? Two Goals

7.1 Two Reasons to Care About f

Why learn f from data? Two distinct motivations:

The two goals

- Prediction: get a good guess $\hat{Y}$ for Y at a new X.
- Inference: understand the *shape* of f — which X matter, in which direction, how strongly.

These goals call for different methods:

- For pure *prediction*, we often don't care what $\hat{f}$ looks like, as long as $\hat{Y}$ is accurate.
- For *inference*, we need $\hat{f}$ to be interpretable — so we can read off what each predictor does.

Sometimes one project asks both questions. The methods we choose reflect which goal matters more.

8. Prediction

8.1 The Prediction Setup

Given a sample of training data (X_i, Y_i) for $i = 1, \ldots, n$, build an estimator $\hat{f}$. Then at a *new* input value $X = x_0$, predict

$$\hat{Y} \;=\; \hat{f}(x_0).$$

Examples:

- Wage: given the worker's education and age, predict their wage.
- Stocks: given yesterday's return and volume, predict whether the market will go up.
- Ads: given a city's ad budgets, predict sales.

In pure prediction mode, $\hat{f}$ is sometimes called a black box: we use it to produce $\hat{Y}$, but we don't insist on understanding the precise form of $\hat{f}$. A random-forest predictor with thousands of trees can be state-of-the-art for $\hat{Y}$ while being humanly uninterpretable — and that's fine for many applications.

8.2 How Accurate Is the Prediction?

$\hat{Y}$ is a random variable (it's a function of the training sample, plus ϵ noise on top). So we need a probabilistic measure of accuracy.

Accuracy measure
The standard measure is the expected squared error:

$$\mathbb{E}\left[(Y - \hat{Y})^2\right].$$

Smaller is better.

Why squared error?

- It's always nonnegative.
- It penalizes large errors more than small ones.
- It connects directly to variance — mathematically clean.

8.2 How Accurate Is the Prediction? 2

This is the same quantity we minimized when we computed the variance in Lectures 1–2: $\text{Var}(Y - \hat{Y})$ measures spread of the prediction errors.

Where this shows up later. Lecture 10 introduces the bias–variance tradeoff for an estimator's mean squared error. For now, just know that $\mathbb{E}[(Y - \hat{Y})^2]$ is *the* target we want to make small.

8.3 Prediction Example: Wages

Setup. We've fit some $\hat{f}$ from training data on 3,000 workers. A new person walks in: 27 years old, 16 years of education. We compute

$$\hat{Y} = \hat{f}(27, 16) = \$52{,}000.$$

Is \$52,000 accurate? On average, across many such people, our predictions hit within some range. Specifically:

- If $\mathbb{E}[(Y - \hat{Y})^2] = 100^2 = 10{,}000$, then the root mean squared prediction error is $\sqrt{10{,}000} = \$100$ on the wage scale.
- Actual industry numbers are much bigger — wage data is noisy and a σ on the order of \$15K is realistic.

Two ways to make our prediction better:

8.3 Prediction Example: Wages 2

- Use a better estimator $\hat{f}$ (more flexible, more data, better features).
- *Cannot* eliminate the irreducible part: even with perfect f, the noise ϵ remains.

Lecture 10 makes this dichotomy formal.

9. Inference

9.1 The Inference Questions

Sometimes we don't just want $\hat{Y}$ — we want to understand f. Typical inference questions:

- Which predictors matter? Often only a handful of the X_j's add useful information about Y. Identifying them is half the battle.
- What is the direction and strength of each association? How do expected wages vary with education, holding everything else fixed?
- Is the relationship linear, or more complicated? Does the relationship between TV ad spending and sales taper off after some level (diminishing returns), or stay linear?
- How confident are we in our answers? Could the apparent relationship be just noise?

Each of these is a question about the *shape and structure* of $\hat{f}$, not about predicting any single $\hat{Y}$.

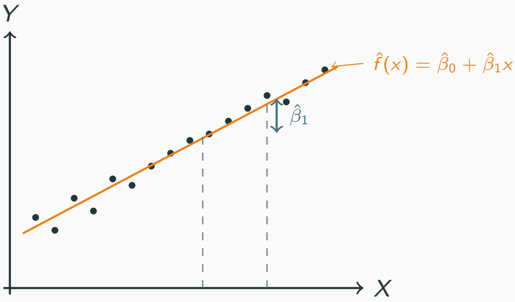
9.2 Inference Needs Interpretable $\hat{f}$

For inference we cannot use $\hat{f}$ as a black box. We need to *look inside* it and read off:

- Which predictors are in the model.
- How $\hat{f}$ depends on each predictor.
- Confidence intervals around the parameters of $\hat{f}$.
- p-values for whether each coefficient is statistically distinct from zero.

This is why most inference work uses interpretable models like linear regression: the coefficients $\hat{\beta}_j$ are directly readable summaries of conditional associations, and the inference machinery from Lecture 4 applies.

9.2 Inference Needs Interpretable $\hat{f}$ 2



The slope $\hat{\beta}_1$ summarizes the linear association between X and Y — immediately readable, with a CI and p-value attached. It is not automatically a causal effect; that interpretation needs additional assumptions or a randomized design. Lecture 7 formalizes the statistical association.

9.3 Prediction vs Inference: A Cheat Sheet

	Prediction	Inference
Goal	accurate $\hat{Y}$	understand f
$\hat{f}$ form	can be black box	must be interpretable
Best methods	flexible, complex	simpler, parametric
Output	predicted $\hat{Y}$	coefficients, CIs, p-values
Examples	forecasting	relationship analysis, A/B tests

Real applications often want both. A bank predicts loan defaults (prediction) but also needs to explain to regulators which variables drive the prediction (inference). The methods that work best for these two goals can be different — one reason this course covers a range of methods.

10. Summary

10.1 Take Away — The Framework

Statistical learning in one slide

- Goal: learn the unknown function f in $Y = f(X) + \epsilon$ from observed (X, Y) pairs.
- ϵ is random noise. Within each observation, $\mathbb{E}[\epsilon_i \mid X_i] = 0$.
- Observed pairs (X_i, Y_i) are independent across i.
- X are inputs / predictors / features (vector of length p). Y is the output / response.
- Regression: Y is quantitative. Classification: Y is qualitative.
- This course covers supervised learning: data contains both X and Y.

10.2 Take Away — The Two Goals

Why estimate f?

- Prediction. Compute $\hat{Y} = \hat{f}(X)$ at a new X. Accuracy measured by $\mathbb{E}[(Y - \hat{Y})^2]$. $\hat{f}$ can be a black box.
- Inference. Understand which X_j are associated with Y, in which direction, and how strongly. Requires $\hat{f}$ to be interpretable. Output: coefficients, confidence intervals, p-values.
- Both. Many real problems combine the two goals — pick methods that serve both.

10.3 Where Each Tool Reappears

Forward pointers

- Linear f — simple linear regression (Lecture 6), inference for the slope (Lecture 7), multiple regression (Lecture 9), regularization (Lecture 10).
- Non-linear / flexible f — non-parametric methods (Lecture 11), trees and random forests (Lecture 14).
- Classification (Y qualitative) — logistic regression (Lecture 12).
- Model assessment — cross-validation and bootstrap (Lecture 13).
- Probability and statistics tools — the entire L1–L4 toolkit (expectation, variance, sampling distributions, CIs, HTs) is what we'll use to evaluate every method that follows.

Next time: simple linear regression — the workhorse model for $f(X) = \beta_0 + \beta_1 X$, where we'll meet our first estimator of f.